

USE CASE DOCUMENT

Document Title	Cyndy's Sweettooth Online Application
Document Owner	Jennifer Francis
Version	1.0
Status	Draft
Date	16/02/2026

BRIEF DESCRIPTION

This use case describes how customers interact with the Online Bakery Application to browse products, place orders, make online payments, and track order status. It also outlines how bakery staff and the admin manage orders, inventory, and reporting.

This use case begins when a customer initiates an order on the platform and ends when payment is successfully processed and the order is confirmed for preparation, or when the transaction fails and is handled appropriately.

ACTORS

- **Customer:** A user who browses products, places orders, and makes online payments.
- **Bakery Staff:** Responsible for preparing orders and updating order status.
- **Admin (Bakery Owner):** Manages products, monitors inventory, and reviews reports.
- **Payment Gateway:** External financial service responsible for securely processing online payments.

PRE-CONDITIONS

- Customer has access to the online application.
- Product catalogue is available in the system.
- Admin and Staff users are registered and logged in.
- Payment gateway integration is active and operational.
- System is online and accessible.

BASIC FLOW

1. Customer accesses the bakery web application.
2. Customer browses available products.
3. Customer selects products and adds them to cart.
4. Customer proceeds to checkout.

5. Customer enters required personal and delivery details.
6. System validates required fields.
7. System displays order summary and total cost.
8. Customer selects payment option.
9. System redirects customer to secure Payment Gateway.
10. Customer enters payment details and confirms payment.
11. Payment Gateway processes transaction.
12. Payment Gateway sends payment confirmation to system.
13. System generates a unique Order ID.
14. System displays payment success and order confirmation message.
15. Inventory levels are automatically updated.
16. Bakery Staff receives notification of paid order.
17. Staff prepares the order and updates order status.
18. Customer tracks order status via the application.

ALTERNATE / EXCEPTION FLOWS:

6a) Missing Required Fields:

If required fields are incomplete, system highlights missing fields and prompts correction.
Process resumes at Step 5.

9a) Payment Gateway Unavailable:

If payment service is unavailable, system displays notification and allows retry.
Customer may retry payment or cancel order.
If cancelled, Use Case Ends.

11a) Payment Failure:

If payment is declined or fails, Payment Gateway notifies system.
System displays payment failure message.
Customer may retry payment.
Process resumes at Step 8.

3a) Out of Stock Item:

If selected product becomes unavailable before checkout, system notifies customer and prevents payment.
Process resumes at Step 2.

16a) Payment Not Confirmed:

If payment confirmation is not received, order status remains "Pending Payment."

Order is not sent for preparation until confirmation is received.

POST-CONDITIONS

- Successful payment results in confirmed order creation.
- Inventory levels are updated automatically.
- Staff can view only paid orders for preparation.
- Customer receives confirmation notification.
- Sales records are updated in reporting dashboard.
- Payment transaction is logged for audit purposes.

If payment fails:

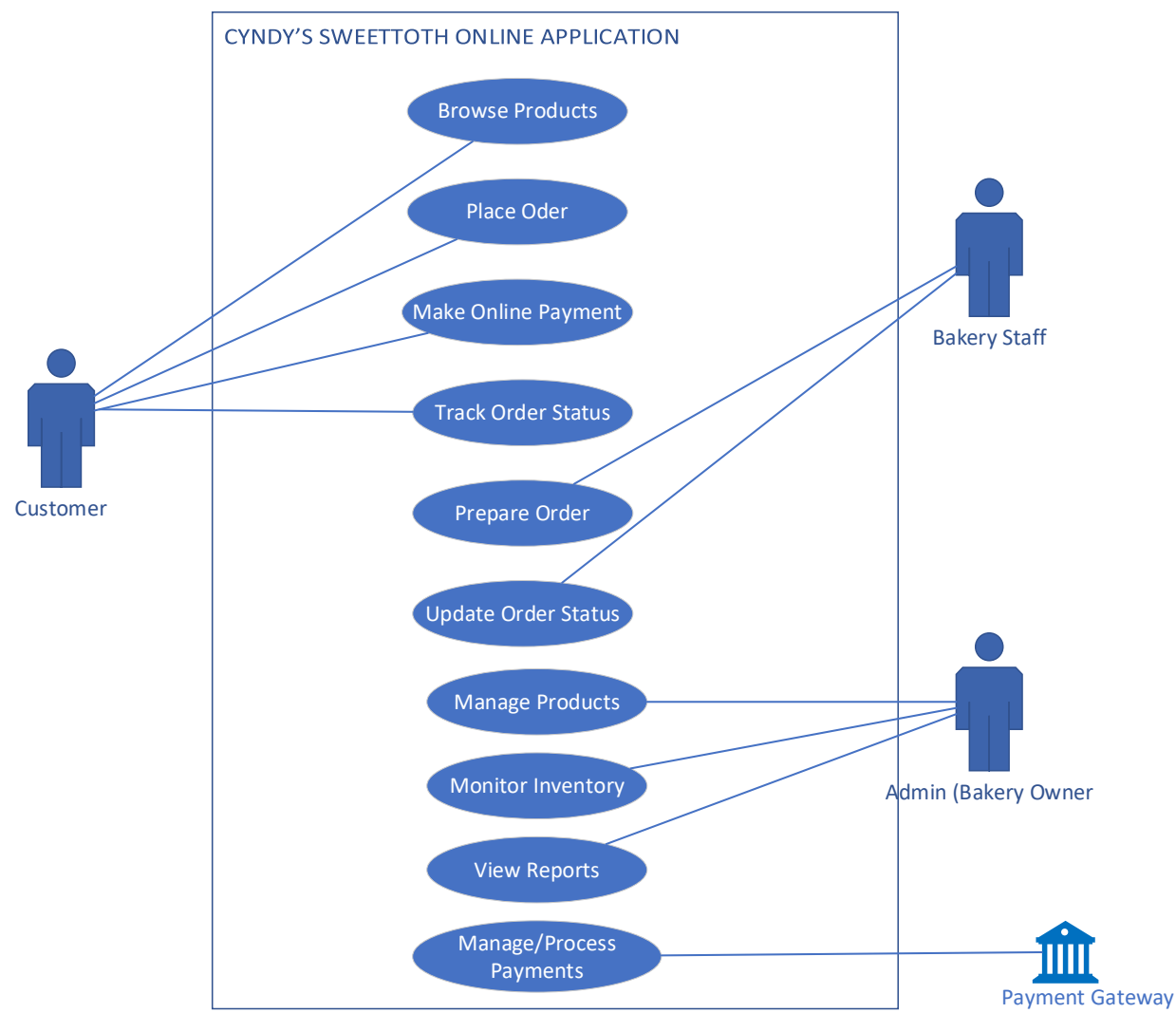
- No order is prepared.
- Inventory remains unchanged.
- System logs failed transaction attempt.

SUPPLEMENTAL REQUIREMENTS

- **Authentication:** Admin and Staff must authenticate with secure credentials.
- **Secure Payment Processing:** All payments must be encrypted and processed via certified Payment Gateway.
- **Role-Based Access Control:** Customers cannot access administrative features.
- **Payment Confirmation Rule:** Orders must not proceed to preparation without verified payment confirmation.
- **Unique Order ID Rule:** Each successful payment generates a unique, timestamped Order ID.
- **Inventory Update Rule:** Inventory must update only after successful payment confirmation.
- **Transaction Logging:** All payment transactions (successful or failed) must be logged.
- **Data Protection:** Customer and payment data must comply with applicable data protection standards.
- **Performance Requirement:** Payment confirmation response time should not exceed 5 seconds under normal conditions.

VISUAL MODEL

The use case may be supported by: UML Use Case Diagram including Payment Gateway actor



Revision History

V.	Date	Author	Description	Status
1.0	16/02/2026	Jennifer Francis-Nkpornwi	Initial Draft	Draft